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Explosive Strength and Speed as Potential Determinants of Success in Youth Figure Skating Competitions

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Abstract: This study aimed to determine the extent to which explosive and reactive strength of the lower limbs, as well as acceleration in running and skating speed, predict figure skating competition results in youth skaters. Talented figure skaters aged 8 to 16 years (n = 211) were divided into two groups: Group 1 (basic novice; age: 10.38 ± 1.10 yrs., height: 1.3889 ± 0.07.26 m, weight: 32.75 ± 4.63 kg) and Group 2 (advanced novice and junior; age: 13.02 ± 1.06 yrs., height: 1.527 ± 0.0818 m, weight: 42.73 ± 7.62 kg). Fifteen variables of physical fitness were measured using tests such as repeated vertical jump, standing long jump, triple jump (right/left leg), 15 m running sprint, and 15 m skating sprint. The competition results were recorded according to the International Skating Union Code of Points. The principal component analysis reduced the fifteen variables to four principal components, explaining 84–87% of the variance. Linear regression models showed that these components significantly explained the variability in the competition scores for both groups. For Group 1, R^2 ranged from 0.298 to 0.425, while for Group 2, it ranged from 0.086 to 0.237. Pearson's R was higher for the mean best score of the season than for the National Championship score. The relative importance of these factors varied across age groups. These findings highlight the need for tailored on- and off-ice training that focuses on locomotion and acceleration for younger skaters and dynamic jump training for older skaters.

Keywords: sport performance; sport training; physical fitness; lower limbs; explosive strength

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1. Introduction

Figure skating is an elegant and captivating sport that seamlessly blends artistic expression, athleticism, and grace on the ice. At the professional and performance levels, figure skating requires intensive training and long-term specialized preparation [1]. To excel, figure skaters need a combination of anthropometric and physical characteristics, including flexibility, strength, power, cardiovascular endurance, agility, speed, coordination, balance, and spatial awareness [2,3]. The ideal combination of these characteristics can vary between skaters, often leading coaches to tailor training programs based on individual strengths and weaknesses of athletes. However, skaters who have a well-rounded combination of these attributes are better positioned for success in figure skating competitions [4]. Additionally, specific physical demands can differ across disciplines such as men's singles, women's singles, pairs, and ice dance [5]. In addition, there are several important areas related to the preparation and performance of figure skating, including motor learning, the quality of mastering specific figure skating skills, the psychological aspects of managing training loads, and the presentation of sports performance. Furthermore, the specifics of pre-adolescent and adolescents sports training, as well as their management, are crucial.

Since 2004, single men and women disciplines of figure skating, pair skating, ice dancing, and synchronized skating have been judged using the International Skating

Union (ISU) judging system, also known as the International Judging System (IJS). The introduction of IJS made the scoring system less vulnerable to manipulation and more objective. Furthermore, significant improvements in skating technique have been observed in competitions following the implementation of IJS. In particular, jumps and spins have become more daring, intricate, and dynamic in competitions [6,7]. Jumps play a critical role in competitive figure skating, as skaters who perform more complicated and complex elements earn higher scores [8]. These visually captivating maneuvers require explosive strength, precision, and grace for successful execution [3]. Lower-limb explosive strength directly affects jump quality, contributing to overall performance and score [9]. Strong lower limbs allow figure skaters to achieve good jump height and length, control rotations, and stabilize landings. Good balance during landing prevents falls, facilitates smooth transitions to the next element, and maximizes points in competitions [10,11].

In elite figure skaters, it has also been documented that the age at which women reach peak performance (20.4 $\pm$ 3.4 years) has decreased and that the number of double jumps has decreased significantly in favor of triple or quadruple jumps [7]. These highlights increase pressure on early specialization, particularly among female skaters, and increase demands on the difficulty of individual elements of figure skating performance, especially the number of jump rotations. Therefore, efficient training with greater overall intensity is required to improve the physical fitness and technical proficiency of figure skaters [7,12].

Although coaches regularly assess figure skater technique in almost every training session, the level and changes in the physical prerequisites for figure skater performance remain somewhat hidden. To address this issue, coaches must focus on other key areas related to sports performance, such as uncovering hidden physical factors, such as strength and power, speed and agility, endurance and stamina, during long-term athlete development. Some cross-sectional and longitudinal studies of figure skaters focus on changes in physical qualities during growth and development and their potential in talent identification. Mostaert et al. [13] identified specific anthropometric, physical, coordination, and ice skating characteristics that discriminate young elite from non-elite figure skaters aged 9 to 12 years. They found no difference between the elite and non-elite groups in maturity status and anthropometric or physical parameters. However, elite skaters scored better than non-elites in the coordination tests to jump sideways and tended to do so in the moving sideways test. Importantly, the finding suggests that no sport-specific motor coordination tests allow discrimination between elite and non-elite skaters and can aid in talent detection and identification among young ice skaters. In a recent study [4], the authors compared the parameters of anthropometric and physical fitness of elite 11- to 16-year-old female and male figure skaters. They concluded that evaluating the relationship between morphological characteristics and motor skills with specific field tests and standardized tests performed at regular intervals in young figure skating athletes can contribute to better selection of athletes, as well as to the effectiveness of their individual training preparation. In another study [5], the physical qualities off the ice in 10- to 33-year-old athletes based on the disciplines and levels of skating were compared. The authors stated that the observed physical qualities varied significantly depending on the discipline and level of skating and that senior and junior skaters tended to exhibit greater speed and strength than novice skaters. These results emphasized the need for strength and conditioning programs that reflect the specific demands of the discipline and the level of skaters. Furthermore, data obtained from off-ice testing can establish normative values for figure skaters. Cruz et al. [2] analyzed longitudinal changes in strength and agility in 9–25-year-old competitive single figure skaters, considering their levels of competition. Skaters remained at the same competitive level on the ice or progressed by one level in consecutive seasons. Surprisingly, the most significant changes occurred in lower-level female skaters who remained at the same level. This finding underlines the importance of early participation in strength and conditioning programs to maximize athleticism before reaching the senior level, where off-ice performance improvements tend to plateau. More specifically, we mean participation in appropriately designed youth resistance training programs that are supervised

and instructed by qualified professionals, which are recommended for aspiring young athletes of this age. These programs must consider biological age, training age, motor skill competency, technical proficiency, and existing strength levels of athletes [14]. We recognize that a successful and effective training process in figure skating is only possible if coaches do not neglect any important areas of sports training and focus on developing the skater as a whole. This involves integrating physical, mental, and emotional aspects to enhance performance. By addressing all these components, a holistic approach aims to create well-rounded athletes who are not only skilled on the ice but also healthy.

The studies mentioned above focused on various fitness parameters and anthropometric characteristics that describe young elite or non-elite figure skaters. However, studies analyzing the association between fitness parameters and success in competitions in figure skaters are missing. As far as we know, only one study [11] investigated the effects of lower-limb strength and endurance on the quality of execution of single jumps (Axel jumps). The findings indicate that muscular strength and endurance have an effect on the training progress and performance of figure skating elements such as jumps. However, none of these studies specifically addresses the question of whether and to what extent fitness parameters influence a figure skater's performance in competition. Therefore, our study aims to determine to what extent explosive strength, reactive strength of the lower limbs, and acceleration in running and skating speed can predict the results of figure skating competitions, as evaluated according to the international rules of the ISU. The hypothesis of the study is that the results of the explosive strength and speed field tests of the lower extremities can predict success in national competitions in elite female and male figure skaters of youth aged 8 to 16 years.

2. Materials and Methods

2.1. Participants

Elite female and male figure skaters aged 8 to 16 years (n = 211) who participated in the Talented Youth Training Project of the Czech Figure Skating Association (CFSA) during the years 2011 to 2024 participated in this study. Figure skaters were divided into two groups by age: Group 1 basic novice (age 8 to 11 years), and Group 2 advanced novice and junior (age 12 to 16 years). The exclusion criteria for motor tests were absence of training in the last two weeks, health problems within six weeks before measurements, and acute health problems at the time of measurement. A priori sample size calculations were performed using the G* Power software (3.1.9.7) after removing outliers to make sure that our sample of skaters was large enough. Our groups contained sufficient numbers of athletes to use the statistical methods chosen and detect meaningful effects: Group 1—85 and Group 2—95 participants (Table 1). The study adhered to the Declaration of Helsinki on the use of human subjects. The methods and procedures applied were carried out in accordance with the stated declaration. The legal guardians of all participants gave their informed written consent for participation in the CFSA Talented Youth Project. Participants and their legal guardians were fully informed about the testing procedures. Written informed consent was obtained from all parents and/or responsible adults.

Table 1. Basic characteristics of Group 1—basic novice and Group 2—advance novice and junior.

	Group 1				**Group 2**			
	Age	**BH**	**BM**	**BMI**	**Age**	**BH**	**BM**	**BMI**
Number of participants	85				95			
Average	10.38	1.38	32.75	17.19	13.02	1.52	42.73	18.49
Standard deviation	1.10	0.07	4.63	1.19	1.06	0.08	7.62	1.77

Note: BH—body height (m), BM—body mass (kg) BMI—body mass index (BMI = BM/BH^2).

2.2. Procedures

Participants were tested at the beginning of the competition period for one day. The day before the test session, participants were not subjected to high-intensity exercises. The off-ice tests were conducted in the indoor sports hall and took approximately 40 min; the indoor ice rink test included an initial ice training of 15 min. Following the basic anthropometric measurement, participants underwent a standardized set of motor tests. The testing was preceded by an individual 10-minute warm-up performed under the supervision of researchers that focused on jumping and running readiness with maximum effort. At the end of the warm-up, the participants were accustomed to the physical test protocol.

2.2.1. Anthropometric Measurements

Body height was measured standing without shoes using a wall-mounted measuring band with an accuracy of 0.01 m. Body mass was measured using a TANITA BC 545 scale (Tanita, Japan) with an accuracy of 0.5 kg.

2.2.2. Tests of Lower-Limb Explosive Strength

Participants underwent standardized lower extremity explosive strength tests in the order listed below. The tests included jumping characterized by a predominant horizontal or vertical component of the forces, and maximal short distance running and skating. A 5 min rest preceded each test.

2.2.3. Repeated Vertical Jump (RVJ) Test

The RVJ test consisted of repeated vertical jumps for 10 s on the FITRO Jumper (FiTRONiC, s.r.o. Bratislava, Bratislava, Slovakia). Each participant, wearing trainers, stood on a jump ergometer with their arms fixed on their sides and held a rubber Thera-Band. The rubber band looped around their waist and arms prevented any accompanying arm movement during the jump. The task was to maximize jump height and minimize ground contact time. Participants performed three 10 s trials of repeated vertical jumps with a 30 s rest between the trials. Before the measured trials, each participant performed one practice trial of approximately 5 s. The height of each jump (h, m) was derived from ft (s) using the formula: h = (g × tf2)/8. The power in the concentric phase of take-off (Pact, W/kg) was calculated using the formula Pact = (g2 × Tf) * (Tc + Tf)/4Tc, where g is the gravitational constant 9.81 m/s2. The RSI was calculated using the equation: RSI = h/Tc [15]. In each trial, the highest Pact achieved in three jumps was averaged. From the trial with the highest average Pact, the average of Tc, h, and RSI of the three jumps were used for the subsequent analysis. The FITRO jumper measures Tc and ft with an accuracy of 1 ms [16], and RSI with reliability ICC = 0.924 [17]. This approach to measuring SSC capability using field-based rebound tests has been shown to be reliable in youth populations [18].

2.2.4. Standing Long Jump (SLJ) Test

SLJ was performed according to the EUROFIT ® standardized physical fitness test battery manual [19]. From a slight crouch position, with feet approximately shoulder-width apart and parallel, each participant squats, leans forward, bumps, and bounces down with a simultaneous swing of their arms forward and jumps as far as possible. The distance of the longest jump of the two trials was used, with an accuracy of 0.01 m. The rest interval between the two trials was 2 min. Before the test trials, each participant performed one practice trial. The SLJ is a test used to assess explosive leg power or the ability to apply force in a horizontal direction [20]. SLJ used in 6- to 12-year-old children was reported to be highly reliable, with a 95% CI for the ICC 0.93–0.95 test–retest [21].

2.2.5. Triple Jump Test with Right/Left Leg (TJR/TJL Test)

This test evaluates an individual's ability to decelerate in contact with the ground for optimal landing and rapidly transition from eccentric to concentric muscle contractions,

i.e., reactive strength [22]. From a standing position on one leg behind the start line, each participant performed three maximal consecutive jumps for distance without pausing between attempts and landing from the last jump on two feet. Each participant performed three trials for each leg, with a two-minute rest interval between the trials. Before the test trials, one practice trial was executed for each leg. The distance from the rebound line to the nearest mark of impact of the foot sole after the third rebound was measured with an accuracy of 0.01 m. The longest of the three attempts was recorded for each leg. The triple jump test showed excellent test–retest reliability with an ICC of 0.94–0.95 [23].

2.2.6. Fifteen-Meter Running Sprint Test

A band was placed around the waist of each participant, with a guide wire connected to the FiTRO speed check device (FiTRONiC, Bratislava, Slovakia). Participants were asked to run from their standing position at their own discretion, with a maximum effort, for a distance of 20 m. They were instructed to start slowing down the run until the wire was detached. Before the measured trial, each participant performed one practice trial. In the case of technical, organizational or task execution problems, the sprint trial was repeated. The total time on the 15 m run (R_t15m, s) and the average velocity at an interval of 0.00–1.00; 1.01–2.00; 2.01- to reach the 15 m distance or 3,00 s of running (R_v_1.s, R_v_2.s, R_v_3.s) were measured The 20 m sprint test showed excellent test–retest reliability of r = 0.90 in children aged 7–11 years [24].

2.2.7. Fifteen-Meter Skating Sprint Test

Participants were in a standing starting position with the back leg skate on the inner edge and the front leg on the skate tooth. A band around the waist of each participant, with a guide wire connected to the FiTRO speed check device (FiTRONiC, Bratislava, Slovakia). Subsequently, participants were asked to drive in a straight line with a maximum effort of 20 m. Each participant performed a trial. Before the measured trial, each participant performed one practice trial. In the case of technical, organizational, or task execution problems, the sprint trial was repeated. We recorded the total time achieved over 15 m (S_t15m) and the average velocity at an interval of 0.00–1.00; 1.01–2.00; 2.01- to reach 15 m distance or 3.00 s (S_v_1.s, S_v_2.s, S_v_3.s). The test–retest reliability for the total 20 m skating time and the mean speed at the first second of skating was found to be 0.989 and 0.968, respectively [25].

2.3. Results of Figure Skating Competitions

From the database of the Czech Figure Skating Association (www.skating-results.live), the results of those skaters who passed in the respective season without limitations and who underwent our standardized tests of explosive strength of the lower limb were selected. For each figure skater, the best three results of Czech figure skating cup competitions that led to a nomination for the National Championships in the given season and the result of the National Championship itself were then selected. The results of the three competitions with the highest ranking in the Czech Figure Skating Cup (SB3) were subsequently averaged. The result of the figure skating competitions according to the ISU Code of Points consists of the sum of points for the short program (SP) and free skate (FS). The total score is the sum of the technical score (Tes) and the program component score (Pcs). In the Official Races and National Championships for young skaters, only free skating is evaluated. Starting from the category of pupils, the short program and the free skate are evaluated separately. In multisegmented competitions (short program and free skate), the score from each segment is added to determine the skater's total score.

2.4. Data Analysis

Data were analyzed using the IBM SPSS software package (version 28.0.1.0). In the first step, outliers of the test scores in the two groups were identified as any observation outside the range (Tukey´s fences): [Q1 − 1.5(Q3 − Q1), Q3 + 1.5(Q3 − Q1)] [26], where

Q1 and Q3 are the lower and upper quartiles. After excluding cases with one or more test scores being outliers, the data for Group 1 (n = 85) and Group 2 (n = 95) were not significantly different from the normal distribution when tested with the Shapiro–Wilk and Kolmogorov–Smirnov tests.

The set of 15 observed variables underwent the principal component analysis (PCA) to reveal significant latent factors. Regarding the fact that we observed strong internal Pearson correlations between variables, we used Oblimin rotation with Kaiser normalization to improve information on factor loadings [27]. The number of significant principal components in the matrix of factor patterns extracted by the PCA was determined by the Kaiser–Guttman criterion [28], which retains principal components with eigenvalues >1. The PCA with the Oblimin rotation and Kaiser normalization procedure was performed on data from both groups.

The data obtained from the PCA for each group was then used in the linear multi regression analysis (MRA). The MRA was used to determine the relationship between IJS scores (dependent variables) and individual PCA factors as representatives of individual items from the lower-limb explosive strength tests. To evaluate the sampling adequacy for each variable in the multi regression model, we used the Kaiser–Meyer–Olkin (KMO) and Bartlett test. The results of the Republic Championship figure skating competitions (RCH) and the best three results during the season (SB3) were used as dependent variables for the MRA. The MRA linear models were constructed separately for each of Group 1 and Group 2 and the IJS score includes seven partial scores: the total score, free skating results (FS_score, FS_Pcs, FS_Tes), and short program results (SP_score, SP_Pcs, SP_Tes). The coefficient of determination (R^2) was used as a measure of the size of the effect, with thresholds following Cohen (1988) for small (0.01), medium (0.09), and large (0.25) effects.

3. Results

The KMO values of 0.807 and 0.84 for the two groups indicate that the data set is suitable for PCA. The Bartlett test showed a significant level $\alpha \leq 0.001$, confirming that the variables were correlated in the population (Table 2A,B).

Table 2. (**A**) Kaiser–Meyer–Olkin (KMO) and Bartlett's test Group 1. (**B**) Kaiser–Meyer–Olkin (KMO) and Bartlett's test Group 2.

(**A**)		
Kaiser–Meyer–Olkin measure of sampling adequacy.		0.84
Bartlett's test of sphericity	Approx. chi-square	1652.04
	Degree of freedom	105
	Significance	≤0.001
(**B**)		
Kaiser–Meyer–Olkin measure of sampling adequacy.		0.807
Bartlett's test of sphericity	Approx. chi-square	1501.018
	Degree of freedom	105
	Significance	≤0.001

The PCA reduced fifteen observed variables to four principal components, which explained 84 to 87% of the variance for both groups.

The PCA model for Group 1 (Table 3) shows that the variables in the sprint run test had the highest loading (correlations) with the first principal component (C1). The second principal component (C2) was highly correlated with the variables of the repeated vertical jump test. The third principal component (C3) was highly correlated with the results of the skating sprint variables and C4 was highly correlated with the results of the standing long jump and triple jump (right/left leg) tests.

Table 3. PCA components loadings of leg strength variables after Oblimin rotation with Kaiser normalization for basic novice (Group 1).

Test	Variables	1	2	3	4
Running sprint test	R_t15 m (s)	**−0.93**		0.501	−0.35
	R_v_2.s (m/s)	**0.892**	0.364	−0.473	0.586
	R_v_3.s (m/s)	**0.81**	0.378	−0.345	0.595
	R_v_1.s (m/s)	**0.78**		−0.333	
Repeated vertical jump test	RSI	0.31	**0.96**	−0.366	0.558
	Pact (W/kg)	0.341	**0.946**	−0.397	0.574
	tc (s)		**−0.847**		
	h (m)	0.444	**0.74**	−0.495	0.662
Skating sprint test	S_v_2.s (m/s)	0.519	0.323	**−0.924**	0.393
	S_t15m (s)	−0.581	−0.339	**0.902**	−0.351
	S_v_3.s (m/s)	0.677	0.316	**−0.883**	0.459
	S_v_1.s (m/s)			**−0.863**	
Jumps in horizontal direction test	TJL (m)	0.444	0.438	−0.355	**0.963**
	TJR (m)	0.403	0.475	−0.378	**0.931**
	SLJ (m)	0.412		−0.313	**0.919**

Note: SLJ (m)—standing long jump, TJR (m)/TJL (m)—triple jump test with right/left leg, tc (s)—contact time, Pact (W/kg)—power in the active phase of the take-off, h (m)—height of the jump, RSI—reactive strength index, R_t15m (s)—running 15 m, R_v_1. s (m/s)—running velocity 1. S, R_v_2. s (m/s)—running velocity 2. S, R_v_3. s (m/s)—running velocity 3. S, S_t15m (s)—skating 15 m, S_v_1. s (m/s)—skating velocity 1.s, S_v_2. s (m/s)—skating velocity 2.s, S_v_3. s (m/s)—skating velocity 3.s. Underline and bolted value are the variables with highest loading with components C1–C4.

In the PCA model for Group 2 (Table 4), the same components were recognized, but the order of the components differed. C1 was highly correlated with the skating sprint test variables, C2 maintained the same order and was highly correlated with the variables from the repeated vertical jump test, C3 was highly correlated with the results of the standing long jump and triple hop jump (right/left leg) tests. The high factor loadings on the fourth principal component (C4) were demonstrated by the results of the running sprint test.

Table 4. PCA components loadings of the leg strength variables after Oblimin rotation with Kaiser normalization for advance novice and junior (Group 2).

Test	Variables	1	2	3	4
Skating sprint test	S_v_2.s (m/s)	**0.929**		0.306	−0.495
	S_t15m (s)	**−0.903**		−0.394	0.537
	S_v_3.s (m/s)	**0.9**		0.394	−0.508
	S_v_1.s (m/s)	**0.873**			−0.331
Repeated vertical jump test	RSI		**0.982**		
	Pact (W/kg)		**0.979**		
	h (m)	0.455	**0.774**	0.421	−0.467
	tc (s)		**−0.768**		
Jumps in horizontal direction	TJL (m)			**0.941**	−0.439
	TJR (m)	0.309		**0.911**	−0.417
	SLJ (m)			**0.899**	−0.464

Table 4. *Cont.*

Test	Variables	1	2	3	4
Running sprint test	R_v_2.s (m/s)	0.393		0.457	**−0.936**
	R_v_3.s (m/s)	0.399		0.479	**−0.915**
	R_t15m (s)	−0.486		−0.415	**0.894**
	R_v_1.s (m/s)	0.591			**−0.697**

Note: SLJ (m)—standing long jump, TJR (m)/TJL (m)—triple jump test with right/left leg, tc (s)—contact time, Pact (W/kg)—power in the active phase of the take-off, h (m)—height of the jump, RSI—reactive strength index, R_t15m (s)—running 15 m, R_v_1. s (m/s)—running velocity 1. S, R_v_2. s (m/s)—running velocity 2. S, R_v_3. s (m/s)—running velocity 3. S, S_t15m (s)—skating 15 m, S_v_1. s (m/s)—skating velocity 1.s, S_v_2. s (m/s)—skating velocity 2.s, S_v_3. s (m/s)—skating velocity 3.s. Underline and bolted value are the variables with highest loading with components C1–C4.

The results of the MRA for Group 1 and Group 2 and each partial competition IJS score are presented in Tables 5 and 6.

Table 5. Statistics of MRA model basic novice Group 1 for results of Republic Championship and the mean of three best season results.

Model	R	R Squa.	Adjust. R Squa.	Std. Error of Estim	F Change	df1	df2	Sig. F Change Model	Durbin–Watson	Signif. Component
RCH_FS_Score	0.578	0.334	0.296	9.206	8.764	4	70	<0.001	1.545	<0.001; C3
RCH_FS_Pcs	0.591	0.35	0.311	4.210	9.136	4	68	<0.001	1.46	<0.001; C3
RCH_FS_Tes	0.546	0.298	0.256	4.590	7.205	4	68	<0.001	1.533	<0.001; C3
SB3_FS_Score	0.625	0.391	0.361	9.358	12.998	4	81	<0.001	1.5	<0.001; C3
SB3_FS_Pcs	0.638	0.407	0.377	4.020	13.873	4	81	<0.001	1.277	<0.001; C3
SB3_FS_Tes	0.652	0.425	0.396	4.243	14.756	4	80	<0.001	0.993	<0.001; C3

Note: RCH—Republic Championship, SB3—three best results, FS—free skates score, Tes—technical score, Pcs—program components score.

Table 6. Statistics of the MRA model advance novice and junior, Group 2 for results of National Championship and the mean of three best season results.

Model	R	R Squa.	Adjust. R Squa.	Std. Error of Estim	F Change	df1	df2	Sig. F Change Model	Durbin–Watson	Signif. Component
RCH_Total sc.	0.439	0.192	0.155	14.473	5.179	4	87	0.001	0.363	<0.001; C3
RCH_SP score	0.431	0.186	0.149	5.379	5.086	4	89	0.001	0.307	<0.001; C3
RCH_SP_Pcs	0.294	0.086	0.042	1.670	1.956	4	83	0.109	0.178	0.027; C3
RCH_SP_Tes	0.431	0.185	0.148	3.185	5.009	4	88	0.001	0.366	<0.001; C3
RCH_FS_Score	0.433	0.188	0.15	10.012	5.023	4	87	0.001	0.469	<0.001; C3
RCH_FS_Pcs	0.392	0.154	0.113	3.954	3.81	4	84	0.007	0.298	<0.001; C3
RCH_FS_Tes	0.388	0.151	0.11	4.903	3.687	4	83	0.008	0.371	<0.001; C3
SB3_Total sc	0.472	0.223	0.193	14.100	7.323	4	102	<0.001	1.635	<0.001; C3
SB3_SP score	0.487	0.237	0.206	4.054	7.69	4	99	<0.001	0.415	<0.001; C3
SB3_SP_Pcs	0.307	0.094	0.056	1.614	2.494	4	96	0.048	0.202	0.048; C3
SB3_SP_Tes	0.434	0.188	0.156	2.644	5.796	4	100	<0.001	0.424	<0.001; C3
SB3_FS_Score	0.473	0.223	0.193	9.980	7.334	4	102	<0.001	1.643	<0.001; C3
SB3_FS_Pcs	0.47	0.221	0.189	3.892	6.888	4	97	<0.001	0.382	<0.001; C3

Note: RCH—Republic Championship, SB3—three best results, total score—SP + FS score. SP—short program score, FS—free skates score, Tes—technical score, Pcs—program components score.

In the MRA models for the basic novice group, Group 1 (Table 5), Pearson's R ranges from 0.546 to 0.652. The R^2 coefficient of determination ranges from R^2 for RCH_FS_Tes = 0.298 (medium effect) for the technical score of free skating, to R^2 RCH_FS_Tes = 0.425 (large effect).

The data in Table 6 show that the MRA models of our components (C1–C4) significantly explain the variability in the IJS score of Group 2 in all cases. The statistical significance for all MRA models for SB3 and RCH IJS scores is $f < 0.001$, except for the models for short program (SP), program component score (RCH_SP_Pcs and SB3_SP_Pcs), where $f < 0.05$. In the MRA models of the advance novice and junior group (Group 2), the Pearson's R ranges from 0.294 to 0.487. The coefficient of determination (R^2) ranges from R^2 for RCH_SP_Pcs = 0.086 (small effect) to R^2 for SB3_SP_score = 0.237 (medium effect). Pearson's R is slightly higher in the regression models for SB3 scores than for RCH scores.

For both groups, basic novice and, advance novice and junior, Pearson's R is slightly higher in the regression models for SB3 scores than for RCH scores. The coefficients of

determination for the IJS scores are slightly higher for the basic novice Group 1 than for the advance novice and junior Group 2.

The MRA models that analyze individual variables of the IJS score demonstrate that the principal components, representing the strength and speed predispositions of young skaters, as determined by field tests, have varying impacts on the overall score and its individual elements. The results of the regression analysis for Group 1 shows that individual components C1–C4 affect the IJS scores of young skaters. Component C3 exerts the greatest influence and is consistently significant in all models, and corresponds to the results of the 15 m skating test. Its influence is statistically significant at the level of $p = 0.001$, affecting both the results of singular events, such as the National Youth Figure Skating Championships, and the recorded results (Table 5).

In the advanced novice and junior category (Group 2), the C3 component, as with Group 1, contributes most significantly to explaining the variance. However, this C3 component, in this instance, is highly correlated with variables representing the results of the TJR, TRL, and SLJ tests (jumps in the horizontal direction) and has a notable effect. Regarding the regression model for the short program components (RCH_SP_Pcs; SB3_SP_Pcs), both for the outcome of the single competition in the National Championship and for the average of the three best races in the season, the effect of C3 is statistically significant at the level of $p = 0.05$. In all other cases, the effect of C3 is significant at the level of $p = 0.01$ (Table 6).

4. Discussion

Previous studies [2,4,5,7,29] in young figure skaters investigated performance on motor ability tests. To the best of our knowledge, our study is the first to assess whether the results of non-specific and specific physical field tests can predict success in competitions in young female and male figure skaters. Vertical jumps, locomotion acceleration, and distance jumps are commonly used in elite athlete training, testing, and monitoring, as well as fitness indicators monitored in this study [30]. The analysis was made based on long-term monitoring of physical parameters such as explosive strength and reactive strength of the lower limbs, acceleration running speed, and acceleration skating speed, and IJS scores in the most important official competitions. The performance level of figure skaters was assessed not only based on the results from the top single event such as the National Championship in the corresponding category, but also using averaged results from the three best placements in competitions that qualified figure skaters for the National Championship. We believe that this approach provided a comprehensive view of the level of performance of young figure skaters included in the Talented Youth Training Project throughout the season.

The data show that our tests assess four neuromuscular qualities derived from identified components: fast SSC capability for vertical jumping, slow SSC capability for horizontal jumping, and fast SSC capability for the acceleration phase in both skating and running. Cross-sectional studies [29,31] indicate that SSC performance improves with age and maturation during childhood and adolescence, which corresponds to our finding that the composition of the main components remains consistent across age groups.

These manifestations of explosive strength are internally uncorrelated (orthogonal), which makes them essential for training control in explosive strength development. According to Suchomel et al. [32], we assumed that muscle strength influences force–time characteristics during various sporting movements, including figure skating [3,10,11,33]. These explosive actions are based on coordinated muscle contractions of the lower extremities during the stretch–shortening cycle (SSC) [11,34–36].

The analyses indicate that locomotion speed and strength, as assessed by field tests in young skaters (8 to 16 years old), are important predictors of long-term performance and success in the National Championships. Higher correlations were found for the SB3 scores, representing the average score of the three best competitions leading up to the National Championship in both categories. This suggests that increased explosive lower-limb strength and locomotion speed allow athletes to effectively practice and execute

more complex elements in training, which are subsequently rewarded with higher scores in competitions. These findings are consistent with the studies by Yordanova [6] and Rauer et al. [7].

In the younger basic novice category (Group 1, 8 to 11 years), the speed of specific locomotion on ice was identified as a significant predictor of performance. This finding underlines the importance of coaching practices that prioritize movement dynamics, which is crucial to execute elements of higher difficulty in free skating. Furthermore, maintaining dynamic balance is essential to master the individual components and combinations of the program, which makes it a determining factor in achieving higher IJS scores in this age Group 1. However, when interpreting the results in figure skaters of this category, it must be considered that even though children of these ages with relatively better speed and explosive strength than their peers may have a natural advantage, this is not definite and although early tests can provide baseline insights into their current abilities and movement quality, they are mainly essential for long-term progression of athletes [37–39].

On the contrary, in the advanced novice and junior group, Group 2 (12 to 16 years), the influence of lower-limb explosive strength parameters, particularly in horizontal jumps (SLJ, TJR/TJL), is more pronounced in explaining the variance in IJS scores. In this age category, the successful execution of jumps and pirouettes, which require substantial explosive lower-limb strength, becomes increasingly important for competitive success. This emphasizes the need for early participation in strength and conditioning programs to maximize athleticism and supports the findings of Cruz et al. [2] and Comuk and Erden [11]. Interestingly, our results suggest that vertical force impulse alone, often viewed by coaches as critical for mastering multi-revolution jumps, is not the sole determinant of success for young figure skaters.

Although limited studies assess the impact of explosive lower-limb strength on competition performance, others, such as Cheng and Gonzalez [40] examined individual elements, particularly jumps. Cabell and Bateman [3] note that the trajectory of a jump is established at take-off, and modifying the shape of the jump arc in air is not possible. Furthermore, longer jumps typically require more horizontal velocity, which can enhance performance in combination jumps. Hirosawa et al. [41] also indicate that jump scoring is influenced by the base value and grade of execution (GOE), linking dynamic force characteristics to successful execution and judges' scoring.

This study has potential limitations. The effect estimates in the model are based on noninterventional and observational methods. Measurement of the explosive power parameters of the lower limbs and locomotion speed was carried out at the beginning of the competition period and may not accurately reflect the current state of the prerequisites for sports performance, especially in view of top competitions such as the National Championships for the relevant age category. More studies should attempt to obtain up-to-date data, especially just before the peak of the figure skating season. Another limitation is the study's focus on the 8–16 age group. Future studies should add information on changes in observed parameters during adolescence, post pubertal age, and adulthood, taking into account the biological development of athletes. Additionally, our study focuses only on a small part of the assumptions of successful athletic performance in figure skating. Future studies should aim to identify other factors that contribute to successful athletic performance and examine how these factors change over time.

5. Conclusions

In conclusion, this study provides novel information on the influence of explosive strength of the lower limbs and acceleration speed of running and skating on competition results in young female figure skaters. The results of our analysis indicate that explosive strength and speed parameters, derived from six motor tests and fifteen monitored variables, can be reduced to four main components, explaining 82–44% of the variance between age groups. Although the composition of these factors is consistent, their relative importance varies in figure skaters of different ages. For skaters aged 8 to 11 years (Group

1), short-distance locomotion speed on ice emerges as the critical factor for scoring. On the contrary, for skaters aged 12 to 16 years (Group 2), explosive strength in horizontal jumps significantly affects competition scores. These findings underline the importance of effective on- and off-ice training that emphasizes locomotion and gradual acceleration for younger skaters, along with developing a dynamic stereotype of jumps in various directions. Future research should explore the long-term effects of these training objectives and their time order on performance outcomes.

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